pmsimstats Codebase Overview

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The `pmsimstats` R package implements Monte Carlo simulation-based power analysis for aggregated N-of-1 clinical trial designs, comparing four trial designs on their ability to detect a biomarker-treatment interaction. Based on Hendrickson et al. (2020), *Frontiers in Digital Health* 2:13.

1 Core R Package Files (R/)

1.1 generateData.R

The data-generating process (DGP). Two exported functions:

- **`generateData()`** – Draws simulated participant data from a multivariate normal distribution. Accepts a pre-built covariance matrix (`cached_sigma`) or builds one via `buildSigma()`. Uses

pre-computed Cholesky factor for efficient MVN draws (`L %*% Z + mu`). Computes outcome columns via vectorized `data.table` operations.

- **`buildSigma()`** – Constructs the 26x26 covariance matrix for a single trial path. Implements:
  - AR(1) within-factor autocorrelation: $\rho^{|t_i - t_j|}$
  - Cross-factor correlations (same-time and different-time with AR(1) decay)
  - Biomarker-BR correlation with exponential decay during off-drug periods: $c.bm \cdot \exp(-\lambda_{cor} \cdot tsd)$
  - Auto-derives $\lambda_{cor} = \ln(2)/\text{carryover_t1half}$
  - Positive definiteness check with verbose warning
  - Pre-computes Cholesky factor for caching
- **`validateParameterGrid()`** – Tests all parameter combinations for PD before simulation. Reports failures upfront.

1.2 generateSimulatedResults.R

The simulation loop. Single exported function:

- **`generateSimulatedResults()`** – Orchestrates the full parameter sweep:
  1. PD pre-validation of all parameter combinations
  2. Sigma matrix caching (builds all unique matrices once)
  3. Parameter grid expansion across designs, response params, baseline params, and model params
  4. Per-parameter-set worker function (`run_one_paramset`) that generates data, fits LME, applies censoring, and collects results
  5. Parallel execution via `furrr::future_map()` with closure-based function passing to workers
  6. Progressive save support for long runs

  Key parameters: `lambda_cor` (correlation decay, default NA = auto), `n_cores` (parallelism, default auto-detect).

1.3 lme_analysis.R

The analysis model. Single exported function:

- **`lme_analysis()`** – Fits a linear mixed-effects model to simulated or actual trial data:
  - Uses `nlme::lme` with `corCAR1(form = ~t|ptID)` for continuous-time AR(1) residual correlation
  - Dynamic formula selection: `Sx ~ bm + t + Dbc + bm:Dbc` when within-subject drug variation exists; `Sx ~ bm + t + bm:t` for open-label
  - `Dbc` is a continuous drug indicator (1 when on drug, exponential decay when off drug)

  - Graceful handling of rank-deficient models (returns NA)
  - Fallback to `lme` without `corCAR1` if optimization fails
  - Extracts interaction coefficient from `nlme` tTable

1.4 carryover_analysis.R

Extended analysis functions for actual trial data:

- **`characterize_carryover()`** – Sweeps candidate carryover half-lives (default 0.1 to 8 weeks), fits `nlme::lme` at each, compares AIC/BIC to identify the best-fitting half-life. Returns drug effect, interaction, AR(1) phi at each half-life.
- **`analyze_trial_extended()`** – Extracts both the main drug effect (`Dbc`) and biomarker interaction (`bm:Dbc`) with 95% CIs. Computes variance components, predicted drug effect as a function of biomarker value, and clinical decision threshold.
- **`print_carryover_summary()`** / **`print_trial_summary()`** – Clinical reporting summaries.

1.5 buildtrialdesign.R

Trial design specification:

- **`buildtrialdesign()`** – Constructs trial paths from intuitive inputs (timepoints, expectancies, on-drug vectors). Computes derived timing variables: `tod` (time on drug), `tsd` (time since discontinuation), `tpb` (time in positive belief).

1.6 utilities.R

Helper functions:

- **`modgompertz()`** – Modified Gompertz function for response trajectories. Passes through origin, asymptotes at `maxr`.
- **`cumulative()`** – Converts interval durations to cumulative time.
- **`reknitsimresults()`** – Recombines progressive save chunks.

1.7 censordata.R

Dropout simulation:

- **`censordata()`** – Two-component dropout: time-dependent (flat rate) and outcome-dependent (biased by symptom change). Implements last-observation-carried-forward censoring.

1.8 plottingfunctions.R

Visualization:

- **`PlotModelingResults()`** – Power heatmaps across 4-dimensional parameter grids with `ggplot2` `geom_tile` and `facet_grid`.
- **`plotfactortrajectories()`** – Factor trajectory plots by trial path.

2 Analysis Scripts

2.1 Figure 4 (`analysis/figure4/`)

Script	Purpose
`01-generate-power-sweep.R`	Figure 4: current code, sequential
`02-generate-power-sweep-original.R`	Figure 4: original commit code
`03-generate-power-by-commit.R`	Figure 4: parameterized by commit
`04-generate-power-head.R`	Figure 4: current code, parallel, c.bm={0,0.25,0.45}
`05-pd-diagnostics.R`	PD failure rate instrumentation
`06-plot-power-heatmaps.R`	Heatmap plotting with provenance

2.2 Figure 5 (`analysis/figure5/`)

Script	Purpose
`01-generate-parameter-sensitivity.R`	Figure 5: revised code
`02-generate-parameter-sensitivity-original.R`	Figure 5: original code

2.3 Extracted Code by Commit

Code from each historical commit extracted for comparison:

```
analysis/scripts/commit_42ac030/   # Initial (publication)
analysis/scripts/commit_8609f12/   # Ron Thomas edits
analysis/scripts/commit_f6ee86d/   # Autocorrelation typo fix
analysis/scripts/commit_f70f86d/   # Carryover added
analysis/scripts/commit_325314f/   # Modified Db (broken)
analysis/scripts/original_R/      # Copy of 42ac030
```

3 White Papers (`docs/pdf/`)

Document	Pages	Content
`04-revised-power-analysis.pdf`	13	Main paper: problems, fixes, validated results
`18-response-parameter-sensitivity.pdf`	6	Response parameter sensitivity before/after
`08-biomarker-correlation-decay.pdf`	10	Pharmacokinetic derivation of lambda_cor
`06-ar1-residual-correlation.pdf`	8	lmer to nlme transition with Type I error validation
`07-positive-definiteness-failures.pdf`	10	PD failure prevalence and causes
`09-carryover-correlation-artifact.pdf`	10	BM-BR correlation artifact analysis
`05-revision-summary.pdf`	12	Revision summary with before/after heatmaps and code

File	Content
`results_core.rda`	Published Figure 4 results (initial commit, 500 reps)
`extracted_bp.rda`	Baseline parameters (BP mean/SD, CAPS mean/SD)
`extracted_rp.rda`	Response parameters (Gompertz max/disp/rate/sd)
`CTdata.rda`	Actual clinical trial data
`results_maxes.rda`	Figure 5A results (response maxima)
`results_rates.rda`	Figure 5B results (response rates)
`results_trajectories.rda`	Factor trajectory data

4 Bundled Data (`data/`)

5 Key Parameters

5.1 Current Configuration (Revised Code)

Parameter	Value	Rationale
Autocorrelation	AR(1), rho = 0.7	PD-feasible, clinically realistic
c.bm	{0, 0.25, 0.45}	Max PD-feasible = 0.45 (OL+BDC)
Carryover t_half	{0, 0.5, 1.0} weeks	Clinically realistic
lambda_cor	ln(2) / t_half (auto)	Pharmacokinetic derivation
Cross-factor (same time)	c.cf1t = 0.2	Unchanged from publication
Cross-factor (diff time)	c.cfct * $\rho^{\lvert t_i - t_j \rvert}$	AR(1) decay applied
Analysis model	nlme::lme + corCAR1	Required for AR(1) Type I error control
PD corrections	0/162 (0%)	All matrices valid

5.2 Publication Configuration (Original Code)

Parameter	Value	Issue
Autocorrelation	CS, rho = 0.8	PD failures at c.bm > 0.25
c.bm	{0, 0.3, 0.6}	0.6 exceeds PD-feasible range
Carryover t_half	{0, 0.1, 0.2} weeks	Too short for genuine effects
BM-BR correlation	Step function	Power drop artifact
Analysis model	lmer (no corCAR1)	Correct under CS
PD corrections	40/162 (24.7%)	Silent, uncontrolled

6 Dependencies

Core: `data.table`, `nlme`, `lme4`, `lmerTest`, `corpcor`, `MASS`, `ggplot2`

Performance: `furrr`, `future` (parallel processing)

Legacy (in DESCRIPTION but used minimally): `svMisc`, `tictoc`, `ggpubr`, `gridExtra`

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Source: ~/prj/alz/10-pmsimstats-ng/pmsimstats-ng/docs/01-codebase-overview.tex